TEXORA · LECTURE SHEET

# Modern Physics

*The quantum revolution in one sheet: blackbody radiation and Planck's hypothesis, the photoelectric effect, Compton scattering, de Broglie waves and uncertainty, the Bohr atom and spectra, X-rays, radioactivity and mass–energy.*

PHY-1203 · PHYSICS-II · TOPIC 6 · LEVEL 1 TERM 2

## 1. Blackbody radiation and the quantum hypothesis

A **blackbody** absorbs all radiation falling on it; heated, it emits a continuous spectrum whose shape depends only on temperature. Two empirical laws organise it: **Wien's law**, $\lambda_{max} T = 2.898 \times 10^{-3}$ m·K (the peak moves to shorter wavelengths as T rises — the Sun, at 5,800 K, peaks at 500 nm, conveniently where eyes are most sensitive), and the **Stefan–Boltzmann law**, emitted power per unit area $= \sigma T^4$ with $\sigma = 5.67 \times 10^{-8}$ W m$^{-2}$ K$^{-4}$. Classical theory predicted ever more emission at short wavelengths — the **ultraviolet catastrophe**. Planck (1900) escaped it by quantising the oscillators: energy exchanges in packets

$$E = hf, \quad h = 6.626 \times 10^{-34} \text{ J·s}$$

and his radiation law then fitted the blackbody spectrum at every temperature — the founding act of quantum physics. Radiation at frequency f comes in **photons**, each of energy hf, momentum $hf/c = h/\lambda$, and zero rest mass.

## 2. The photoelectric effect

Light striking a metal ejects electrons — but only above a threshold frequency $f_0$, however bright the light; above threshold, electron energy grows with frequency, not intensity, and emission is instantaneous. Wave theory predicts none of this; Einstein's photon picture predicts all of it:

$$hf = \varphi + KE_{max}, \quad \varphi = hf_0 \text{ (the work function)}$$

**WORKED EXAMPLE — SODIUM, Φ = 2.28 EV**

Use hc = 1,240 eV·nm. Violet light 400 nm: photon energy = 1,240/400 = 3.10 eV, so $KE_{max}$ = 3.10 − 2.28 = **0.82 eV** (stopping potential 0.82 V). Threshold: $\lambda_0$ = 1,240/2.28 = **544 nm** — redder light frees nothing, at any intensity.

## 3. The Compton effect

Compton (1923) scattered X-rays off electrons and found the scattered wavelength *lengthened* by an angle-dependent amount — exactly as if a photon (particle with momentum h/λ) bounced elastically off an electron:

$$\Delta\lambda = (h/m_e\,c)(1 - \cos\theta), \quad h/(m_e\,c) = 2.43 \times 10^{-12}\ \text{m} = 0.0243\ \text{Å}$$

At θ = 90° the shift is one Compton wavelength, 0.0243 Å, independent of the target material — particle-like light, beyond reasonable doubt.

## 4. Matter waves and uncertainty

De Broglie (1924) proposed the symmetry: every particle carries a wave of wavelength

$$\lambda = h/p = h/(mv)$$

An electron accelerated from rest through V volts has λ (in ångströms) = 12.27/√V — at 100 V, λ = 1.23 Å, atomic-spacing size, which is why electron beams diffract from crystals (Davisson–Germer) and why electron microscopes outresolve light microscopes. Heisenberg's **uncertainty principle** sets the price of measurement: $\Delta x\,\Delta p \geq \hbar/2$. Confining an electron to an atom ($\Delta x \approx 10^{-10}$ m) forces $\Delta p \geq 5.3 \times 10^{-25}$ kg m s$^{-1}$, i.e. $\Delta v \approx 5.8 \times 10^{5}$ m s$^{-1}$ — atoms cannot have still electrons, which is why matter does not collapse.

## 5. The Bohr atom and spectra

Bohr quantised the hydrogen atom: electrons occupy orbits with angular momentum mvr = nħ, radiating only in jumps. Energies and the spectral law:

$$E_n = -13.6/n^2\ \text{eV}, \quad 1/\lambda = R\,(1/n_1^2 - 1/n_2^2), \quad R = 1.097 \times 10^{7}\ \text{m}^{-1}$$

Series: Lyman (to n = 1, ultraviolet), Balmer (to n = 2, visible), Paschen (to n = 3, infrared). Balmer's red line Hα (3 → 2) carries 13.6 × (1/4 − 1/9) = 1.89 eV, λ = 656 nm — the red glow of hydrogen lamps and nebulae. Bohr's model fails for multi-electron atoms, but its energy levels survive intact in full quantum mechanics.

## 6. X-rays

Fast electrons slammed into a metal target produce X-rays two ways: **bremsstrahlung** (braking radiation, a continuous spectrum with a sharp short-wavelength cut-off set by the full electron energy, λ_min = hc/eV — the Duane–Hunt law: a 40 kV tube cuts off at 0.31 Å) and **characteristic lines** when

inner-shell vacancies refill (Moseley's law ties their frequencies to atomic number — the modern basis of the periodic table's ordering).

## 7. Radioactivity and mass–energy

Unstable nuclei decay randomly but predictably in aggregate: $N = N_0 e^{(-\lambda t)}$, with half-life $T_{½} = \ln 2/\lambda$ and activity $A = \lambda N$ (becquerels). Alpha decay sheds $^4$He nuclei, beta decay converts neutrons ↔ protons (with the neutrino balancing the books), gamma decay sheds surplus energy. Einstein's **$E = mc^2$** prices all of it: mass is frozen energy, and the **mass defect** of a nucleus is its binding energy. For the deuteron: proton + neutron = 1.007276 + 1.008665 = 2.015941 u; the deuteron weighs 2.013553 u; the missing 0.002388 u × 931.5 MeV/u = **2.22 MeV** of binding — the same 2.22 MeV gamma emitted when a proton and neutron fuse. Fission and fusion are arithmetic on this curve: nuclei rearrange toward iron, the most bound nucleus, releasing the difference.

## 8. Key points

- Planck $E = hf$ resolves the ultraviolet catastrophe; Wien $\lambda_{max} T$ and Stefan–Boltzmann $\sigma T^4$ organise blackbody emission.
- Photoelectric: $hf = \varphi + KE_{max}$; threshold frequency, linear KE–f graph, instant emission — photons, not waves.
- Compton shift $\Delta\lambda = (h/m_e c)(1 - \cos\theta)$ proves photons carry momentum $h/\lambda$.
- De Broglie $\lambda = h/p$; electron at 100 V → 1.23 Å; uncertainty $\Delta x \Delta p \geq \hbar/2$ stabilises atoms.
- Bohr: $E_n = -13.6/n^2$ eV; Rydberg formula predicts every hydrogen series.
- Radioactive decay $N = N_0 e^{(-\lambda t)}$, $T_{½} = \ln 2/\lambda$; mass defect × 931.5 MeV/u = binding energy.

## 9. Practice

Answer in your own words; brief model answers follow.

1. A metal has work function 2.0 eV. Light of 300 nm falls on it. Find the maximum kinetic energy of the photoelectrons and the stopping potential.
2. X-rays of wavelength 1.00 Å scatter from a target at 90°. Find the scattered wavelength.
3. Calculate the de Broglie wavelength of an electron accelerated through 400 V, and of a 0.10 kg ball moving at 20 m $s^{-1}$. Comment on the comparison.
4. Find the wavelength of the Lyman-α line (n = 2 → 1) of hydrogen and the region of the spectrum it lies in.

5. A radioactive sample has a half-life of 5 days. What fraction remains after 15 days? If the initial activity is 800 Bq, what is it then?
6. An X-ray tube runs at 50 kV. Find its minimum wavelength. Why is this cut-off independent of the target material while the characteristic lines are not?

## Answer key

1. Photon energy = 1,240/300 = 4.13 eV. KE_max = 4.13 − 2.0 = 2.13 eV; stopping potential = 2.13 V.
2. $\Delta\lambda$ = 0.0243 Å at 90°, so $\lambda'$ = 1.0243 Å. (The electron recoils with the balance of energy and momentum.)
3. Electron: $\lambda = 12.27/\sqrt{400} = 12.27/20 = 0.61$ Å — atomic scale, so diffraction and microscopes work. Ball: $\lambda = h/(mv) = 6.626 \times 10^{-34}/(0.10 \times 20) = 3.3 \times 10^{-34}$ m — 24 orders below any grating spacing; its wave nature is unobservable, and classical mechanics reigns. The difference is purely h being small on human scales.
4. $\Delta E = 13.6 \times (1 - 1/4) = 10.2$ eV; $\lambda = 1{,}240/10.2 = 121.6$ nm — ultraviolet (Lyman series lies wholly in the UV).
5. 15 days = 3 half-lives → $(1/2)^3 = 1/8$ remains. Activity scales the same way: 800/8 = 100 Bq.
6. $\lambda_{min} = hc/eV = 1{,}240$ eV·nm/50,000 eV = 0.0248 nm = 0.248 Å. The cut-off is set by energy bookkeeping alone (one electron's whole KE into one photon), so the target is irrelevant; the characteristic lines are fingerprints of the target's own inner energy levels, so they depend entirely on the material.

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